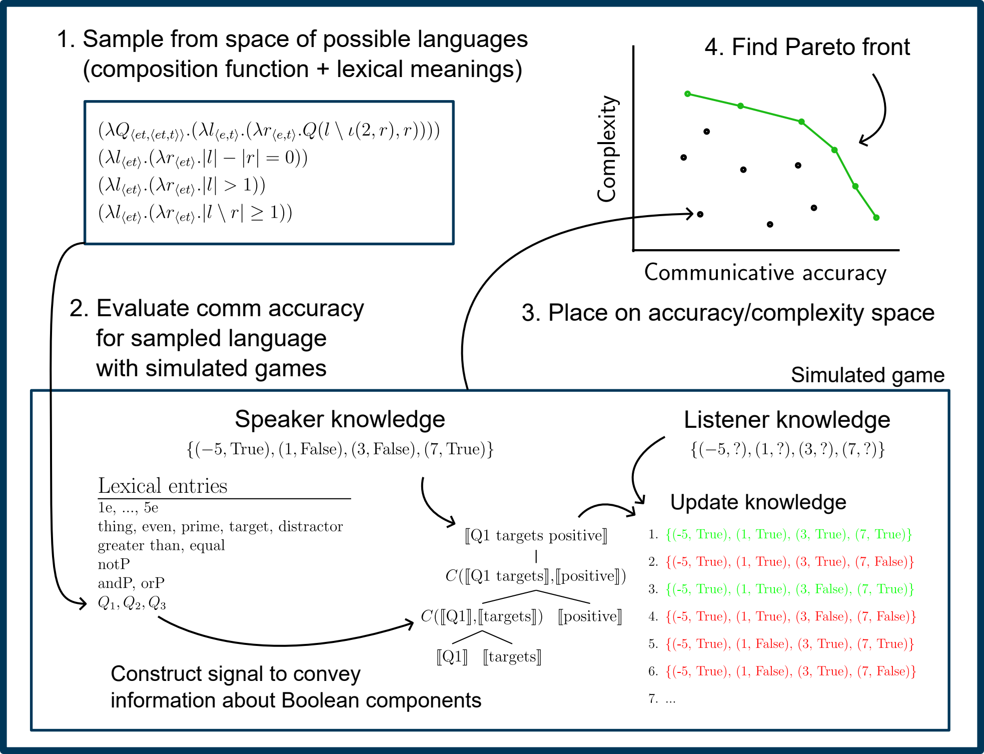

1. Sample from space of possible languages (composition function + lexical meanings)
$(\lambda Q_{\langle et,\langle et,t\rangle\rangle}.(\lambda l_{\langle e,t\rangle}.(\lambda r_{\langle e,t\rangle}.Q(l \setminus \iota(2,r),r))))$
$(\lambda l_{\langle et\rangle}.(\lambda r_{\langle et\rangle}.|l| - |r| = 0))$
$(\lambda l_{\langle et\rangle}.(\lambda r_{\langle et\rangle}.|l| > 1))$
$(\lambda l_{\langle et\rangle}.(\lambda r_{\langle et\rangle}.|l \setminus r| \geq 1))$
4. Find Pareto front
Complexity
Communicative accuracy
2. Evaluate comm accuracy for sampled language with simulated games
3. Place on accuracy/complexity space
Simulated game
Speaker knowledge
$\{(-5, \text{True}), (1, \text{False}), (3, \text{False}), (7, \text{True})\}$
Listener knowledge
$\{(-5, ?), (1, ?), (3, ?), (7, ?)\}$
Lexical entries
1e, ..., 5e
thing, even, prime, target, distractor
greater than, equal
notP
andP, orP
Q_1, Q_2, Q_3
⟦Q1 targets positive⟧
C(⟦Q1 targets⟧,⟦positive⟧)
C(⟦Q1⟧,⟦targets⟧)
⟦positive⟧
⟦Q1⟧
⟦targets⟧
Update knowledge
1. {(-5, True), (1, True), (3, True), (7, True)}
2. {(-5, True), (1, True), (3, True), (7, False)}
3. {(-5, True), (1, True), (3, False), (7, True)}
4. {(-5, True), (1, True), (3, False), (7, False)}
5. {(-5, True), (1, False), (3, True), (7, True)}
6. {(-5, True), (1, False), (3, True), (7, False)}
7. ...
Construct signal to convey information about Boolean components